SCFL: Symbolic Context Field Learning

A Formal Architecture for Symbol-First Language Acquisition,
Node Emergence, Contextual Memory, and Semantic Field Anchoring

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Abstract

This document formalizes a symbol-first learning architecture in which semantic structure emerges from the interaction of atomic symbols, contextual clustering, verified nodes, and a small set of foundational semantic fields. The architecture is designed as an alternative to token-first and embedding-first paradigms. Instead of beginning with a fixed vocabulary and learning dense representations through large-scale gradient descent, the proposed system begins from a universal symbol substrate, constructs symbolic co-occurrence graphs, forms candidate nodes through stable clustering, verifies them as semantic units through contextual evidence, and then anchors them into six foundational semantic fields through dynamic resonance.

The central claim is that meaning should be built upward from structure rather than imposed downward from a fixed vocabulary. At the lowest level, the system operates over single symbols, including characters, ideographs, punctuation, emoji, mathematical notation, and any atomic Unicode unit. At the next level, recurrent symbolic structures condense into nodes. These nodes do not become stable concepts merely because they appear often, but because they recur across diverse and informative contexts. Contextual frequency, contextual diversity, and surprise are therefore treated as core learning variables. Verified nodes are subsequently mapped onto a field substrate, where their field weights evolve over time as the system learns more. In this process, semantic development resembles child language acquisition: repeated exposure, expanding contexts, gradual stabilization, and later abstraction.

The framework further defines the cross-layer weights linking symbol to symbol, symbol to node, node to context, and node to field. It introduces node verification criteria, field balancing, capacity constraints, entropy regularization, and dynamic update rules. The result is a formal specification for a semantic learning substrate in which meaning emerges from symbolic interaction, contextual memory, and field organization rather than from opaque parameter-heavy embedding spaces alone.

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1. Motivation

Most contemporary language models begin from a tokenization scheme that compresses raw text into pre-defined units and then maps those units into vector spaces learned through large-scale optimization. This approach is effective, but it hides several foundational questions. Where does semantic structure come from before token embeddings stabilize? Why should a vocabulary be assumed before the system has learned the structure of language from the ground up? Why should meaning be treated as a property of tokens alone rather than as something that emerges from repeated contextual interaction?

The present framework answers these questions with a symbol-first architecture. The system does not begin from words. It begins from atomic symbols. It does not assume meaning in advance. It allows recurrent symbolic organization to generate candidate semantic units. It does not treat repetition alone as learning. It treats learning as the accumulation of distinct and informative contextual mappings. It does not assign fixed meaning vectors once and for all. It gradually anchors verified nodes into a small set of semantic fields and updates those anchors as experience grows.

Core thesis. Meaning should emerge from symbol interaction, context accumulation, node stabilization, and semantic field anchoring. A language system should not require a fixed vocabulary before it can begin to learn structure.

2. System Overview

The full architecture proceeds through the following stages:

Raw Input → Symbol Layer → Symbol Co-occurrence Graph → Candidate Node Formation → Contextual Le

Operationally, the architecture may be summarized as:

1. universal symbol intake;
2. storage and indexing by IPOD data ontology;
3. co-occurrence learning over local windows;
4. symbol clustering into candidate nodes;
5. contextual strengthening of nodes;
6. verification of node stability;
7. resonance-based anchoring into six semantic fields;
8. dynamic field balancing as new nodes arrive.

3. Foundational Definitions

3.1. Atomic Symbol

Definition 3.1 (Atomic Symbol)**.** *An* atomic symbol *is the smallest representational unit admitted by the system. It may be a character, punctuation mark, ideograph, emoji, mathematical sign, or any other normalized single-symbol unit. The global symbol alphabet is*

$$\Sigma = \{s_1, s_2, \ldots, s_n\}.$$

3.2. Symbol Sequence

Given an input text object X, after normalization it is represented as a sequence

$$X = (s_1, s_2, \ldots, s_T),$$

where each $s_i \in \Sigma$.

3.3. Candidate Node

Definition 3.2 (Candidate Node)**.** *A* candidate node *is a stable cluster of symbols recurrently associated through symbolic co-occurrence and contextual recurrence. Nodes are denoted by*

$$n \in \mathcal{N}.$$

3.4. Context

Definition 3.3 (Context)**.** *A* context *of a symbol or node is the structured neighborhood in which it appears. At the symbol layer, context is given by local symbol windows. At the node layer, context is given by co-occurring nodes, neighboring structures, and larger discourse environments.*

3.5. Semantic Field

Definition 3.4 (Semantic Field)**.** *A* semantic field *is a foundational semantic axis to which verified nodes may be anchored. The field set is*

$$\mathcal{F} = \{F_1, F_2, F_3, F_4, F_5, F_6\}.$$

These fields are primitive axes, not final categories of meaning.

4. Layer 0: Universal Symbol Intake

4.1. Tokenizer Principle

The tokenizer operates at the level of normalized single symbols rather than pre-defined word pieces. Formally, let the tokenizer be a mapping

$$\tau : X \mapsto (s_1, s_2, \ldots, s_T).$$

The normalization subroutine may be written:

$$s_i = \text{Normalize}(c_i),$$

where c_i is the raw Unicode codepoint or raw textual symbol.

Normalization may include:

- Unicode normalization (for example NFC or NFKC),
- canonical symbol mapping,
- optional symbol collapsing,
- script-specific preservation rules.

4.2. Why Symbol-First

This layer is universal across languages and symbol systems. It does not require:

- a pre-built vocabulary,
- BPE merges,
- language-specific token assumptions,
- word boundary assumptions.

Principle of Universal Symbol Intake

Any symbol system should be processable by the substrate before lexical meaning is known. Lexical structure must emerge from symbol dynamics, not be required in advance.

5. IPOD Data Ontology at the Symbol Layer

At the first learning layer, each symbol s is stored under an IPOD-like ontology:

```
Symbol/
    s/
        I/
        P/
        O/
        D/
```

5.1. I: Input Occurrences

$$I(s) = \{p_1, p_2, \ldots, p_m\},$$

where each p_i is an occurrence position of symbol s in the corpus.

5.2. P: Process Context

For a window radius w, the local context of symbol s_i is

$$C_i(s) = \{s_{i-w}, \dots, s_{i-1}, s_{i+1}, \dots, s_{i+w}\}.$$

The collection of all such windows yields:

$$P(s) = \{C_1(s), C_2(s), \dots\}.$$

5.3. O: Output Associations

The output layer stores likely continuations or follow-up symbols:

$$O(s) = \{s_{i+1}, s_{i+2}, \dots\},$$

or equivalently conditional structure such as

$$P(s_j \mid s_i).$$

5.4. D: Data Metadata

This layer stores symbol-level statistics such as:

- raw frequency,
- normalized frequency,
- entropy of neighbors,
- distribution over contexts,
- candidate cluster participation.

6. Symbol Co-occurrence Learning

6.1. Local Co-occurrence Matrix

Let the sequence be $X = (s_1, \dots, s_T)$. For a fixed context window size w, define the local context of s_i:

$$C(s_i) = \{s_{i-w}, \dots, s_{i-1}, s_{i+1}, \dots, s_{i+w}\}.$$

The symbol co-occurrence count is:

$$M_{ab} = \#(s_a \text{ occurs near } s_b).$$

Conditional probability:

$$P(s_b \mid s_a) = \frac{M_{ab}}{\sum_k M_{ak}}.$$

6.2. Pointwise Mutual Information

To capture meaningful attraction beyond raw frequency, define:

$$\mathrm{PMI}(s_a, s_b) = \log \frac{P(s_a, s_b)}{P(s_a)P(s_b)}.$$

High PMI indicates that two symbols occur together more strongly than chance would predict.

6.3. Symbol Graph

The symbol layer induces a weighted graph:

$$\mathcal{G}_\Sigma = (\Sigma, \mathcal{E}_\Sigma),$$

where edges are weighted by

$$w_{ab}^{(\Sigma)} = \mathrm{PMI}(s_a, s_b).$$

Remark 6.1. *Raw degree alone is insufficient. High-frequency symbols such as spaces or common letters may have large degree without strong semantic force. PMI-like weighting is therefore necessary.*

7. Candidate Node Formation

7.1. From Symbol Associations to Candidate Nodes

A candidate node is formed from a subset of symbols whose internal association is sufficiently strong. Let

$$n = \{s_{i_1}, s_{i_2}, \ldots, s_{i_r}\}.$$

Node candidacy depends on symbolic cohesion. One possible cohesion score is:

$$\mathrm{Coh}(n) = \frac{1}{|E(n)|} \sum_{(u,v)\in E(n)} \mathrm{PMI}(u, v),$$

where $E(n)$ is the set of internal symbol-pair edges within the candidate cluster.

A candidate node is admitted if:

$$\mathrm{Coh}(n) > \theta_{\mathrm{coh}}.$$

7.2. Stability Against Combinatorial Explosion

Because symbol clustering can produce vast numbers of accidental fragments, a further stability criterion is needed. Define:

$$\sigma(n) = \frac{\mathrm{freq}(n)^\alpha \ \mathrm{Coh}(n)^\beta}{\mathrm{H}(\mathrm{context}(n)) + \varepsilon},$$

where $\varepsilon > 0$ avoids division by zero.

Candidate nodes are kept only if:

$$\sigma(n) > \theta_\sigma.$$

This suppresses spurious clusters and preserves stable symbolic patterns.

8. Three-Layer Learning and Cross-Layer Weights

The architecture has three essential learning layers:

1. Symbol layer,
2. Node (or word-like cluster) layer,
3. Context layer.

These layers require cross-layer weights.

8.1. Weight Type I: Symbol-to-Symbol Attraction

This is given by:

$$w_{ab}^{(\Sigma)} = \mathrm{PMI}(s_a, s_b).$$

It measures the local attraction between symbols.

8.2. Weight Type II: Symbol-to-Node Binding

Let $\beta_{a,j}$ denote the contribution of symbol s_a to node n_j. A normalized binding can be written:

$$\beta_{a,j} = \frac{\mathrm{PMI}(s_a, n_j)}{\sum_k \mathrm{PMI}(s_a, n_k)}.$$

This answers:

How much does symbol s_a belong to node n_j ?

8.3. Weight Type III: Node-to-Context Association

Let the context class set be

$$\mathcal{C} = \{c_1, c_2, \ldots, c_p\}.$$

Define:

$$\gamma_{j,k} = P(c_k \mid n_j).$$

This answers:

In which context does node n_j usually appear?

8.4. Composed Link Across Layers

A symbol may induce context participation through node mediation:

$$\omega_{a,k} = \sum_j \beta_{a,j}\gamma_{j,k}.$$

Thus:

$$s_a \to n_j \to c_k.$$

This creates a continuous learning chain across the three layers.

9. Node Strength: Why More Links Is Not Enough

A node with many links is not necessarily stronger in the semantic sense. A useful node-strength measure must combine multiple factors.

9.1. Degree

Raw degree:

$$\deg(i) = \sum_j A_{ij}.$$

This is insufficient by itself.

9.2. PMI Network Weight

The total informational strength of node i:

$$W_i = \sum_j \mathrm{PMI}(i, j).$$

9.3. Clustering Coefficient

A node is stronger when its neighbors are also tightly linked:

$$C_i = \frac{2E_i}{k_i(k_i - 1)},$$

where k_i is degree and E_i counts neighbor-neighbor edges.

9.4. Temporal Stability

One possible stability score is:

$$S_i = \frac{\mathrm{freq}(i)}{\mathrm{Var}(\mathrm{context}(i)) + \varepsilon}.$$

9.5. Field Resonance Contribution

After field anchoring exists, a node also derives strength from field resonance:

$$R_i = \sum_{k=1}^{6} f_k \cdot \mathrm{sim}(n_i, F_k).$$

9.6. Composite Node Strength

A combined strength score is:

$$\mathrm{Strength}(i) = \alpha W_i + \beta C_i + \gamma S_i + \delta R_i.$$

Node strength principle. A node is strong not merely because it has many links, but because its links are informative, coherent, stable, and field-relevant.

10. Contextual Learning: Memory Strength as Contextual Frequency

10.1. Why Raw Frequency Is Not Enough

If only raw repetition is counted, highly frequent function symbols dominate:

$$\text{freq}(n) \quad \text{alone is insufficient.}$$

10.2. Contextual Frequency

Define the set of distinct contexts of node n as:

$$\mathcal{C}(n) = \{c_1, c_2, \ldots, c_q\}.$$

Then contextual frequency is:

$$\text{CF}(n) = |\mathcal{C}(n)|.$$

This measures how many distinct contexts the node has mapped into.

10.3. Context Diversity

Let $p(c)$ be the empirical probability of context c for node n. Then:

$$D(n) = -\sum_{c \in \mathcal{C}(n)} p(c) \log p(c).$$

This captures whether the node appears in many truly different contexts rather than repetitive copies of the same one.

10.4. Surprise

A node may also learn faster when it appears in unexpected contexts:

$$\text{Surprise}(n, c) = -\log P(n \mid c).$$

High surprise increases learning pressure because the context challenges prior expectation.

10.5. Learning Strength

We therefore define node learning strength as:

$$L(n) = \text{freq}(n)^{\alpha} \cdot \text{CF}(n)^{\beta} \cdot D(n)^{\gamma} \cdot \left(1 + \overline{\text{Surprise}}(n)\right)^{\delta},$$

where $\overline{\text{Surprise}}(n)$ is the average surprise of node n over its observed contexts.

Child-Like Learning Principle

A symbol or node is not remembered strongly merely because it repeats often, but because it repeats across many different, informative, and surprising contexts.

11. Node Verification

Nodes should only be anchored into semantic fields once they are sufficiently stable.

Definition 11.1 (Verified Node)**.** *A node n is* verified *if its learning strength and structural coherence exceed prescribed thresholds:*

$$\mathrm{Verify}(n) = 1 \quad \Longleftrightarrow \quad L(n) > \theta_L \quad \textit{and} \quad \mathrm{Coh}(n) > \theta_{coh}.$$

Only verified nodes are promoted into the semantic field layer.

This gives the architecture a developmental logic:

$$\text{symbol} \to \text{pattern} \to \text{node} \to \text{verified concept}.$$

12. Six Semantic Fields

After node verification, the system introduces six foundational semantic fields:

$$\mathcal{F} = \{F_1, F_2, F_3, F_4, F_5, F_6\}.$$

These fields are initially empty:

$$\mathcal{N}_{F_k}(0) = \varnothing, \qquad k = 1, \ldots, 6.$$

The six fields function as primitive semantic axes rather than final semantic labels. Their exact interpretation may remain partially open during early learning and become more structured over time.

Remark 12.1. *The fields should be thought of as seed axes of semantic organization, not as fixed taxonomic boxes that predetermine all later meaning.*

13. Field Anchoring Through Resonance

13.1. Field Resonance

For a verified node n and field F_k, define the field resonance:

$$\mathrm{Res}(n, F_k).$$

A simple formulation is:

$$\mathrm{Res}(n, F_k) = \frac{1}{|\mathcal{C}(n)|} \sum_{c \in \mathcal{C}(n)} \mathrm{sim}(c, \mathrm{proto}(F_k)),$$

where $\mathrm{proto}(F_k)$ is the current field prototype.

13.2. Field Weights

Node-to-field weights are normalized:

$$w_{n,k} = \frac{\mathrm{Res}(n, F_k)}{\sum_{j=1}^{6} \mathrm{Res}(n, F_j)}.$$

Thus each verified node receives a field vector:

$$\mathbf{W}_n = [w_{n,1}, w_{n,2}, w_{n,3}, w_{n,4}, w_{n,5}, w_{n,6}], \qquad \sum_{k=1}^{6} w_{n,k} = 1.$$

13.3. Learning-Weighted Resonance

Because node maturity matters, we may strengthen resonance by node learning strength:

$$\mathrm{Res}'(n, F_k) = L(n) \cdot \mathrm{Res}(n, F_k).$$

Then the normalized field weight becomes:

$$w_{n,k} = \frac{\mathrm{Res}'(n, F_k)}{\sum_{j=1}^{6} \mathrm{Res}'(n, F_j)}.$$

This ensures that more mature nodes exert stronger semantic pull.

14. Field Balancing and Dynamic Update

As new nodes are verified and anchored, the field system must remain balanced.

14.1. Total Field Load

The total load of field F_k is:

$$S_k = \sum_{n \in \mathcal{N}_{\mathrm{ver}}} w_{n,k}.$$

14.2. Global Normalization

A normalized node-field contribution may be defined as:

$$\widetilde{w}_{n,k} = \frac{w_{n,k}}{S_k + \varepsilon}.$$

This prevents any one field from absorbing all semantic mass.

14.3. Field Capacity

Each field may also have a capacity:

$$K_k > 0.$$

When S_k approaches or exceeds K_k, the field becomes saturated. One possible penalty is:

$$\widehat{w}_{n,k} = w_{n,k} \cdot \exp\left(-\lambda \max\left(0, \frac{S_k - K_k}{K_k}\right)\right).$$

14.4. Soft Field Competition

An alternative dynamic assignment is:

$$w_{n,k} = \frac{\exp(\mathrm{Res}(n, F_k)/T)}{\sum_{j=1}^{6} \exp(\mathrm{Res}(n, F_j)/T)},$$

where $T > 0$ is a temperature parameter.

14.5. Incremental Update Rule

As the node learns more contexts over time, its field vector drifts:

$$\mathbf{W}_n^{(t+1)} = (1 - \eta)\mathbf{W}_n^{(t)} + \eta\, \mathbf{R}_n^{(t)},$$

where $\mathbf{R}_n^{(t)}$ is the current normalized resonance vector and $0 < \eta \leq 1$.

This models developmental semantic growth.

15. Avoiding Semantic Collapse

Without regularization, the system risks semantic collapse, where too many nodes converge onto one field.

15.1. Field Entropy

For field F_k, define:

$$H(F_k) = -\sum_n w_{n,k} \log w_{n,k}.$$

15.2. Entropy Regularization

A regularization penalty may encourage field diversity:

$$\mathcal{L}_{\text{field}} = -\lambda \sum_{k=1}^{6} H(F_k).$$

Maximizing field entropy discourages trivial collapse.

Field Diversity Principle

A viable semantic field system must preserve differentiated organization. Fields should attract nodes, but not annihilate distinction by collapsing all nodes into one semantic basin.

16. End-to-End Learning Pipeline

The full pipeline may now be written as:

$$X \to \tau(X) \to \mathcal{G}_\Sigma \to \mathcal{N}_{\text{cand}} \to L(n) \to \text{Verify}(n) \to \mathbf{W}_n \to \text{field organization.}$$

Expanded version:

$$\begin{aligned}
X &\mapsto (s_1, \ldots, s_T), \\
(s_1, \ldots, s_T) &\mapsto \mathcal{G}_\Sigma \text{ via co-occurrence learning}, \\
\mathcal{G}_\Sigma &\mapsto \mathcal{N}_{\text{cand}} \text{ via clustering}, \\
n &\mapsto L(n) \text{ via frequency, contextual frequency, diversity, surprise}, \\
n &\mapsto \text{Verify}(n), \\
n &\mapsto \mathbf{W}_n \text{ via field resonance}, \\
\{\mathbf{W}_n\} &\mapsto \text{balanced semantic field structure.}
\end{aligned}$$

17. Interpretation as Child-Like Learning

The architecture is developmentally motivated.

A child does not begin with a fixed dictionary. A child first encounters:

symbols and patterns → repeated fragments → stabilized words → meaning across contexts → abstraction.

This is mirrored in the present architecture:

$$\text{symbol} \to \text{cluster} \to \text{node} \to \text{verified concept} \to \text{field anchoring.}$$

Repeated occurrence alone does not create strong memory. Repeated occurrence across many distinct contexts creates stable concept formation.

18. Contrast with Token-First LLM Architectures

Aspect	**Typical LLM**	**SCFL**
Starting unit	token / subword	atomic symbol
Vocabulary assumption	required early	not required early
Meaning formation	embedding-first	structure-first
Main learning driver	parameter optimization	context accumulation
Interpretability	low	higher
Node stability source	hidden in training weights	explicit verification rules
Semantic organization	implicit latent space	explicit field anchoring
Personalization potential	external or later	native via field dynamics

19. Formal Summary of the Missing Element

The missing element in the broader project is a learning substrate in which:

1. symbols are universal primitives,

2. nodes emerge from symbol dynamics,
3. memory strength depends on contextual recurrence,
4. concepts become stable only after verification,
5. verified nodes are anchored to a semantic field substrate,
6. field structure evolves dynamically as experience grows.

In compact form:

$$\text{meaning} \neq \text{fixed token vector.}$$

Instead:

$$\text{meaning} = \text{symbol interaction} + \text{context accumulation} + \text{node stabilization} + \text{field anchoring.}$$

20. Conclusion

This document formalized a complete symbol-first semantic learning architecture under the name *SCFL: Symbolic Context Field Learning*. The framework begins from a universal symbol substrate, constructs co-occurrence relations, forms candidate nodes through symbolic cohesion, strengthens nodes through contextual recurrence, verifies them as concepts through explicit thresholds, and finally anchors them into a six-field semantic substrate. The field system then evolves dynamically as new verified nodes are acquired.

The decisive conceptual shift is that meaning is not assumed at the vocabulary level. Meaning emerges from repeated structural interaction. Raw repetition is not enough; contextual frequency, diversity, and surprise determine learning strength. Verified nodes do not merely exist as isolated units; they become part of a broader semantic field geometry. In this way, the system develops in a manner closer to developmental learning than to static vocabulary lookup.

The resulting architecture provides a principled substrate for later stages of semantic reasoning, dynamic context fields, personalized field resonance, family-word retrieval, and intention-sensitive response generation. As such, this document should be read as the formal specification of a missing foundational layer: the layer in which language is first learned as structure before it is used as thought.

Appendix A: Compact Equation Set

$$\Sigma = \{s_1, \ldots, s_n\}$$

$$M_{ab} = \#(s_a \text{ near } s_b)$$

$$\mathrm{PMI}(s_a, s_b) = \log \frac{P(s_a, s_b)}{P(s_a)P(s_b)}$$

$$\mathrm{Coh}(n) = \frac{1}{|E(n)|} \sum_{(u,v)\in E(n)} \mathrm{PMI}(u,v)$$

$$\mathrm{CF}(n) = |\mathcal{C}(n)|$$

$$D(n) = -\sum_{c\in\mathcal{C}(n)} p(c)\log p(c)$$

$$\mathrm{Surprise}(n,c) = -\log P(n \mid c)$$

$$L(n) = \mathrm{freq}(n)^{\alpha} \cdot \mathrm{CF}(n)^{\beta} \cdot D(n)^{\gamma} \cdot (1 + \overline{\mathrm{Surprise}}(n))^{\delta}$$

$$\mathrm{Verify}(n) = 1 \iff L(n) > \theta_L \quad \text{and} \quad \mathrm{Coh}(n) > \theta_{\mathrm{coh}}$$

$$w_{n,k} = \frac{\mathrm{Res}(n,F_k)}{\sum_{j=1}^{6} \mathrm{Res}(n,F_j)}$$

$$S_k = \sum_n w_{n,k}$$

$$H(F_k) = -\sum_n w_{n,k} \log w_{n,k}$$

Appendix B: Suggested Layer Names

Layer	**Role**
Layer 0	Universal symbol tokenizer and normalization
Layer 1	Symbol co-occurrence graph
Layer 2	Candidate node formation from symbol clusters
Layer 3	Contextual learning and node-strength accumulation
Layer 4	Verified semantic node layer
Layer 5	Six-field semantic anchoring and balancing
Layer 6	Higher semantic organization / later DCF integration